

# *FFF Printing Fundamentals*

*Part 3: Dimensional Accuracy*

# Dimensional Accuracy

- Be careful modeling parts with exact dimensions.
- Why?
  - 3D prints are imperfect.
  - Errors compound when you assemble printed parts together.
- Where do these errors come from?
- And how do we fix it?



# Dimensional Accuracy

- Sources of error include but are not limited to:
  - Filaments that shrink.
  - Shapes that easily warp.
  - Inconsistent extrusions.
  - Racked gantry (skewed axes).
  - Failing cooling fan.
  - Incorrect belt tension.
  - Backlash, slop, or loose parts.
  - Moving parts that are worn down.
  - Sloppy slicer settings.
  - Thermal expansion (*discussed in Part 3*).
  - Vibrations (*discussed in Part 3*).

# Modeling Calibration

- $(\text{Printed size}) = (\text{Size}) \times (\text{Scale factor}) + (\text{Constant offset})$ .
- Simplified model.
  - Assume fully functional motion system.
  - Apply to X and Y dimensions.
- **Scale factor.** Error changes with size.
  - Part shrinkage.
- **Constant offset.** Error is always the same.
  - Extrusion inconsistencies. Over and under-extruding.
  - Surface roughness. Wet filament, additives like carbon fiber, over-extruding.



([xkcd.com](http://xkcd.com))

# Fixing Scale Factor

- **Adjust part scaling in the slicer to fix scale factor.**
  - Scale XY size, or
  - Dedicated shrinkage compensation setting.
- *Example:* If you part shrinks by 1%, scale the size to 101%.
- When do you need part scaling?
  - Large parts over 6 inches, 150 mm.
  - Filaments that shrink, like ABS.
- I ignore scale factor for small projects that use PLA and PETG.

# Fixing Constant Offset

- **Add clearance (air gap) between parts that touch to fix constant offset.**
- Very important for projects with assembled pieces!
  - All sizes.
  - All filaments.
- Best to add clearance while designing 3D objects.
  - Greatest flexibility. Independently adjust each surface.
- Or try adding negative XY offset in slicer.
  - Equally offsets all vertical surfaces.

# Adding XY Clearance

- How much XY clearance? It depends ...
- **Start with 0.25 mm.** Offset each side by half, 0.125 mm.
- *Example:* Making a hole for a bolt.
  - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
  - Add 0.4 - 0.8 mm of clearance for a loose fit.

# Adding Z Clearance

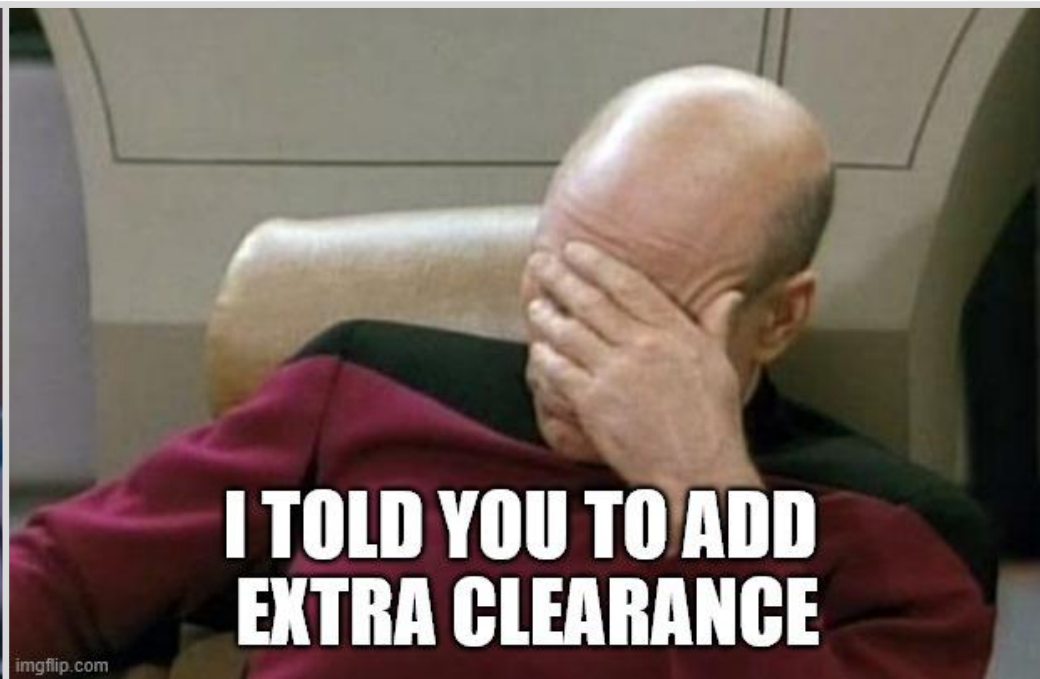
- How much Z clearance? It depends ...
- **Start with one layer or 0.1 mm.**
- When should you increase clearance?
  - Sagging extrusions.
  - Rough bottom surface from supports.
  - High and low spots on print bed.
  - Tilted print bed.



# Summary

If your project has assembled parts then ...

- You sometimes need scale factor.
- You **always** need constant offset.



# Calibration Prints

- Calibration cubes.
  - Labels for X and Y-axis.
  - Tells you constant offset for outside dimensions.
  - Simple but still useful.
- Clearance test by 3DMakerNoob.
  - Print-in-place clearance/tolerance.
  - Alters the gap around discs.
  - Feel which pieces move freely.



# Calibration Prints

- Califlower by Vector3D.
  - Paid model.
  - Includes walkthrough, code, and calculator.
  - Inner dimensions.
  - Outer dimensions.
  - Shrinkage.
  - Skew (if axes are not 90°).
- Calibration video by CNC Kitchen.

